La Scuola d'Italia / Dr. Huson / IB Math 12
19 October 2026

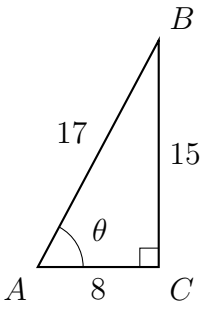
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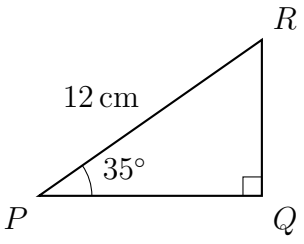
2.1 Classwork: Right-Angled Triangle Trigonometry

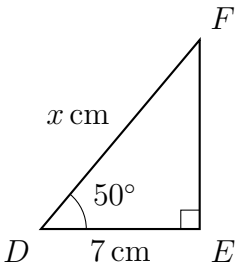
Set your calculator to degree mode. Give answers correct to three significant figures.

1. For an acute angle θ in a right-angled triangle,

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$







   (a) In triangle ABC, write down the length of the hypotenuse, the side opposite θ, and the side adjacent to θ.

   (b) Write down $\sin\theta$, $\cos\theta$ and $\tan\theta$ as fractions.

   (c) Find the size of angle θ.

   (d) In triangle PQR, find the length QR.

   (e) In triangle DEF, find the value of x.

2. A ladder 6 m long leans against a vertical wall. The foot of the ladder is on horizontal ground, 1.8 m from the base of the wall. It may help to draw a diagram.

   (a) Find the angle between the ladder and the ground.

   (b) Find how far up the wall the ladder reaches.

   A vertical tower BT stands on horizontal ground. From a point P on the ground, 40 m from B, the angle of elevation of the top T is 28°. A car is parked at point C, on the other side of the tower, 60 m from B.

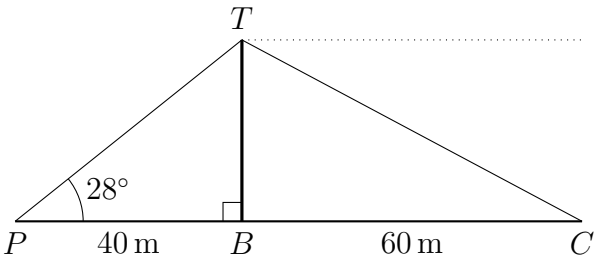


diagram not to scale

   (c) Find the height of the tower.

   (d) Find the distance PT.

   (e) Find the angle of depression of the car C from the top of the tower, T.